StrataWave Validation Pack

Technical validation summary for the hybrid 1D site-response platform. This pack is an external deliverable built from existing repository evidence.

Generated UTC	2026-03-23T13:32:39.461950+00:00
Core verdict	Calisiyor
Parity verdict	Kismi
Release verdict	Kismi
Overall verdict	Kismi

1. Scope

StrataWave currently behaves as a hybrid platform: DEEPSOIL-like workflow, OpenSees-backed effective-stress execution, and native MKZ/GQH analysis paths.

2. Claims supported by this pack

- Native linear, EQL, and nonlinear analysis paths execute end-to-end.
- The OpenSees effective-stress adapter executes and captures diagnostics.
- Darendeli-based MKZ/GQH calibration is wired into the repository.
- DEEPSOIL parity tooling exists, but parity remains partial.
- The web UI exposes wizard, results, parity, and confidence surfaces.

3. Validation snapshot

Smoke pack	4/4 passed
OpenSees parity benchmark	ran=0, skipped=1
Scientific confidence rows	5

4. Status matrix

- Done: Native linear/EQL/nonlinear; OpenSees adapter; Darendeli calibration; DEEPSOIL-like UI; benchmark harness (Current)
- Partial: DEEPSOIL parity, release-signoff runner in local environment, visual appendix coverage (Near-term)
- Pending: Broader published-reference matrix and native full effective-stress solver (Next)
- Out-of-v1: Full native u-p solver and complete DEEPSOIL project import/export (Later)

5. Evidence inventory

- Smoke pack: native example bundle passed 4/4 cases.
- OpenSees evidence: effective-stress smoke logs and diagnostics exist.
- Calibration evidence: Darendeli-based MKZ/GQH wiring is present.
- Parity evidence: compare engine, manifest samples, and release wiring exist.
- UI evidence: wizard, results, parity, and confidence panels are live.

Full artifact paths and notes are preserved in the Markdown and JSON manifest files.

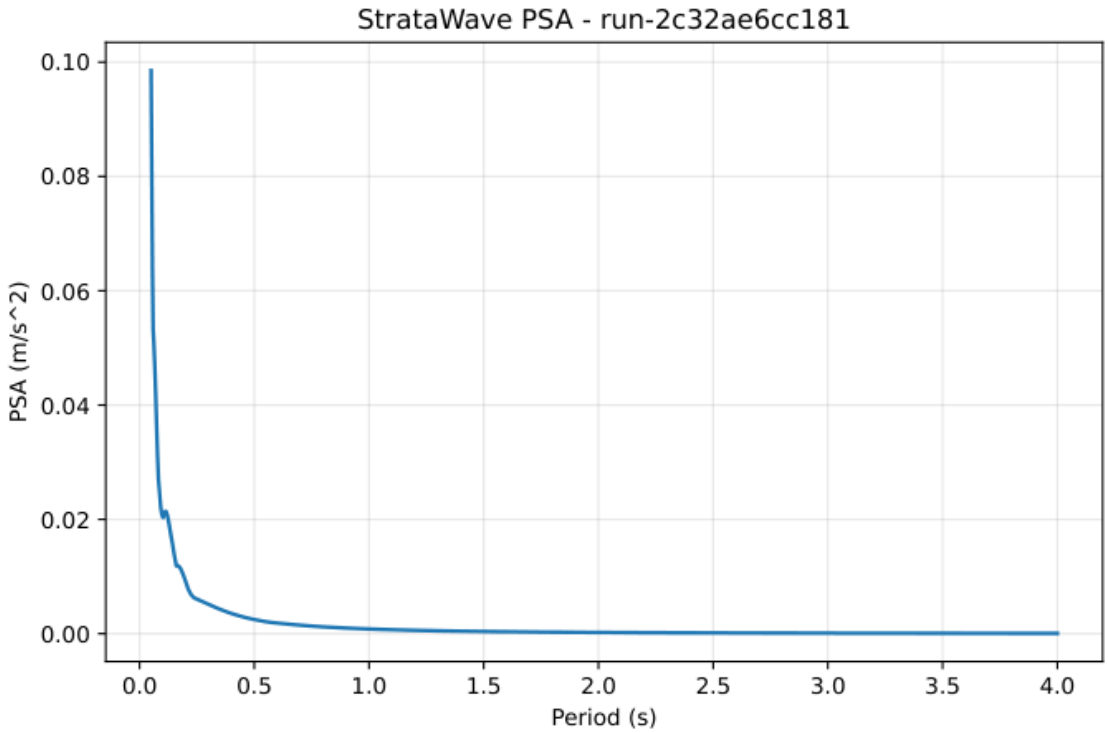
6. Scientific confidence

Canonical release-scientific signoff source of truth.

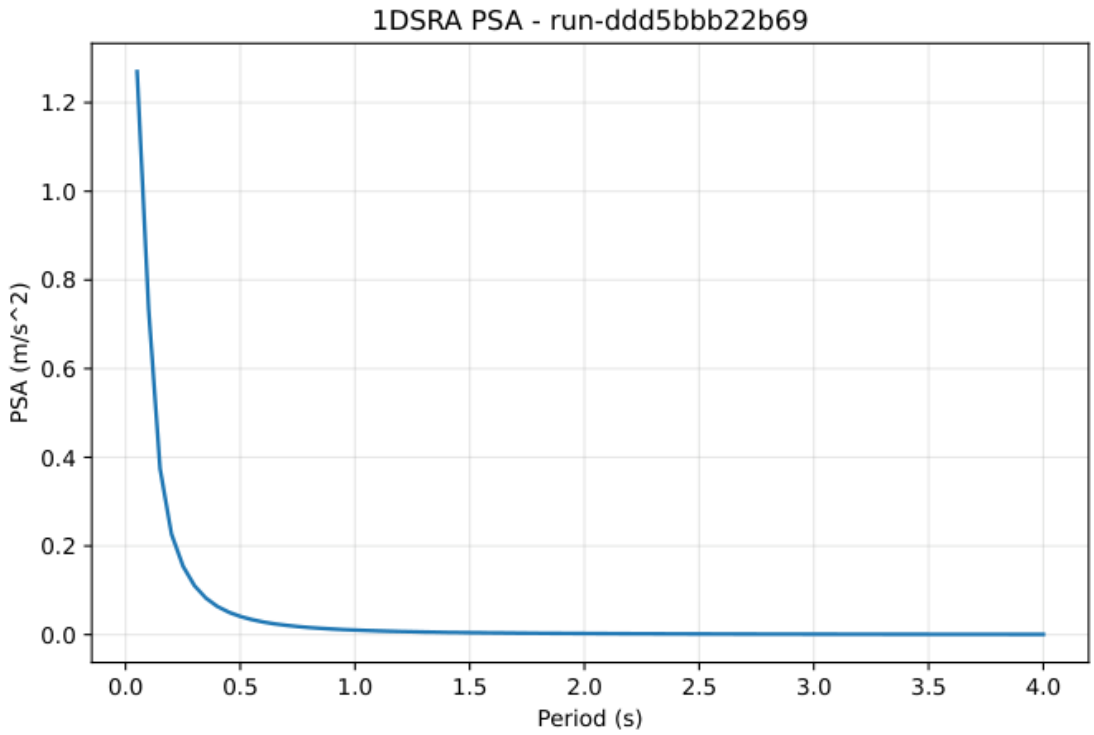
Tracked suites	5
Reference file	SCIENTIFIC_CONFIDENCE_MATRIX.md

7. Visual appendix previews

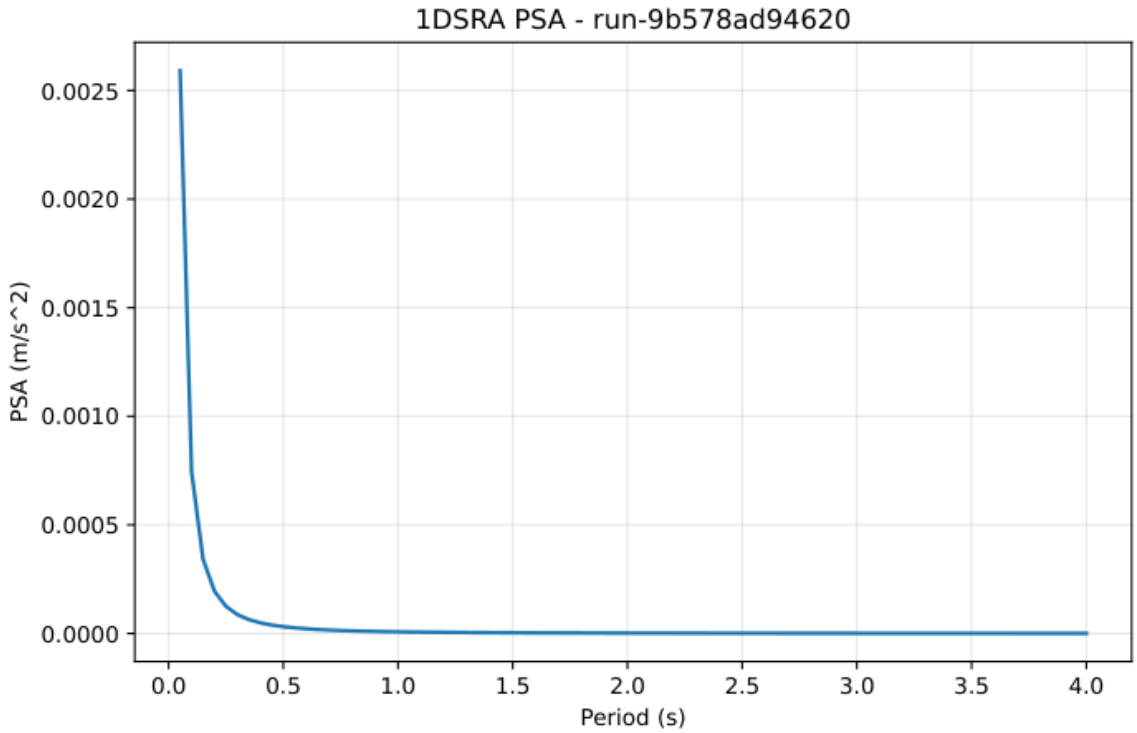
Preview source: run-2c32ae6cc181



Preview source: run-ddd5bbb22b69



Preview source: run-9b578ad94620

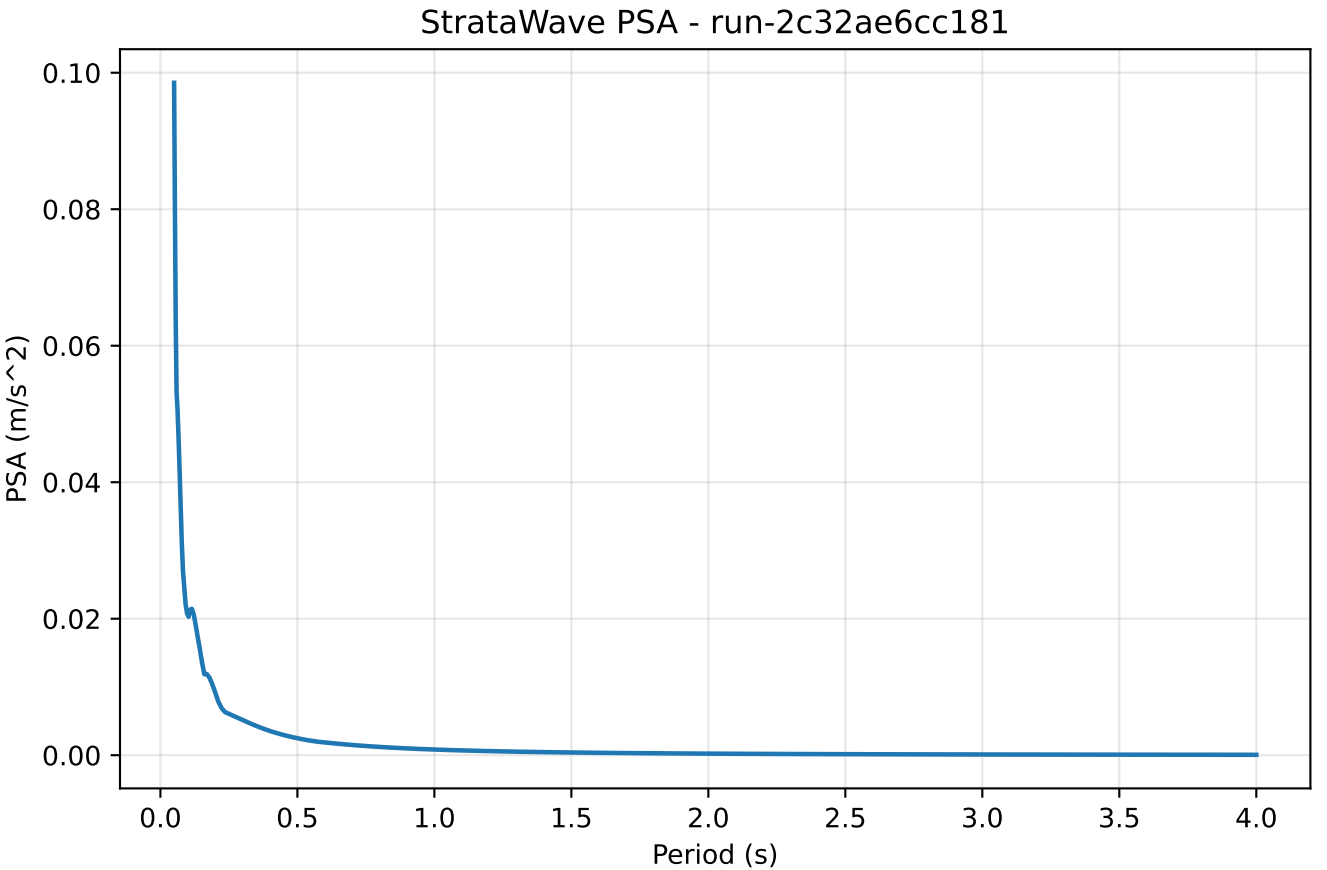


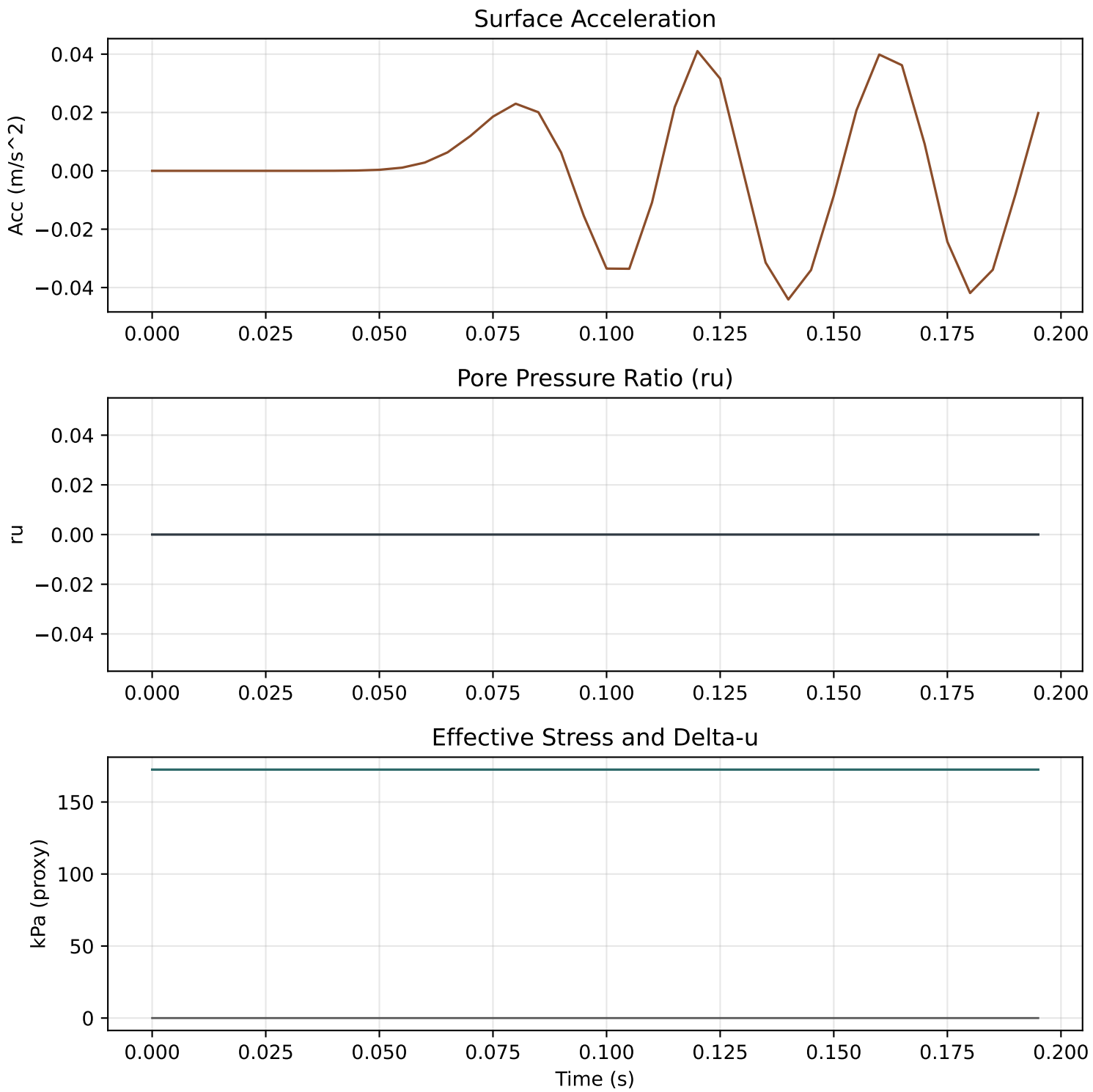
8. Honest limitations

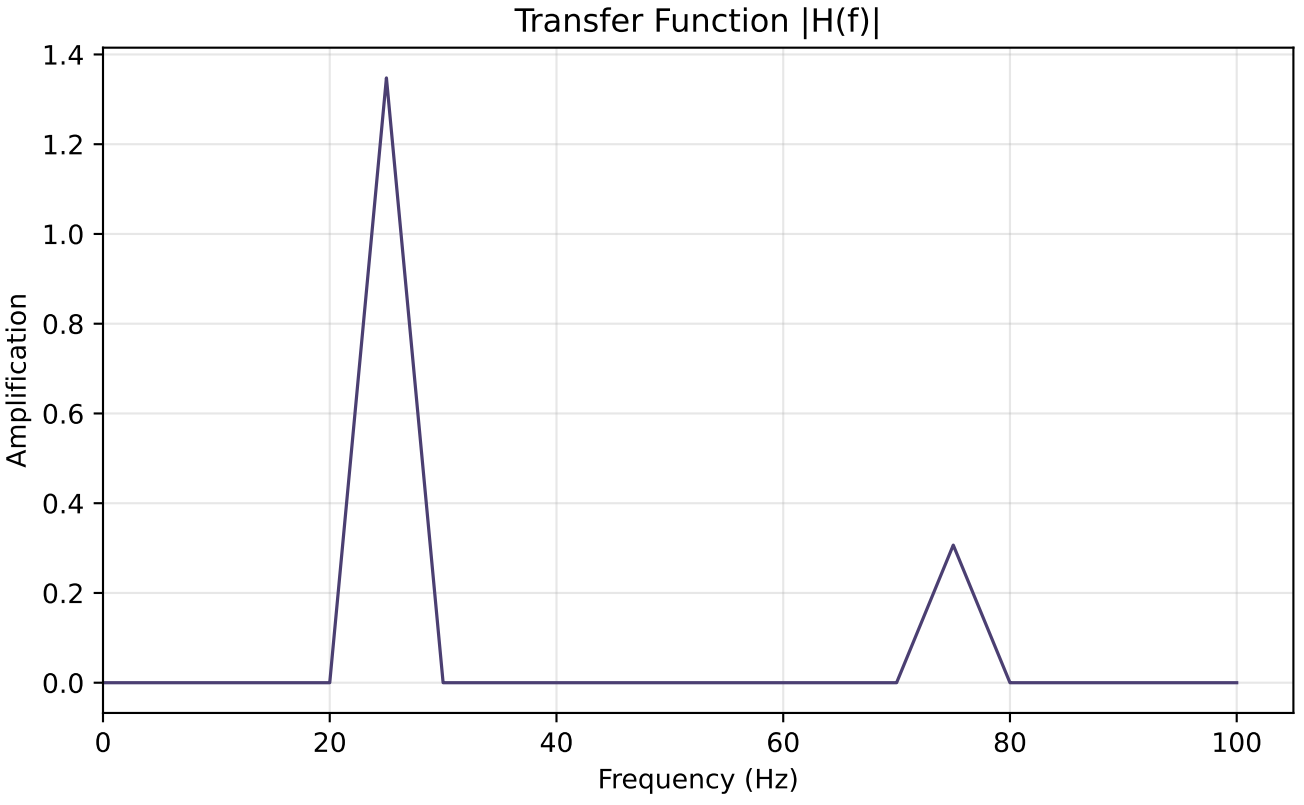
- This pack does not claim full DEEPSOIL parity.
- The native full effective-stress solver is still pending.
- Dedicated OpenSees runner coverage is environment-dependent and policy-gated.

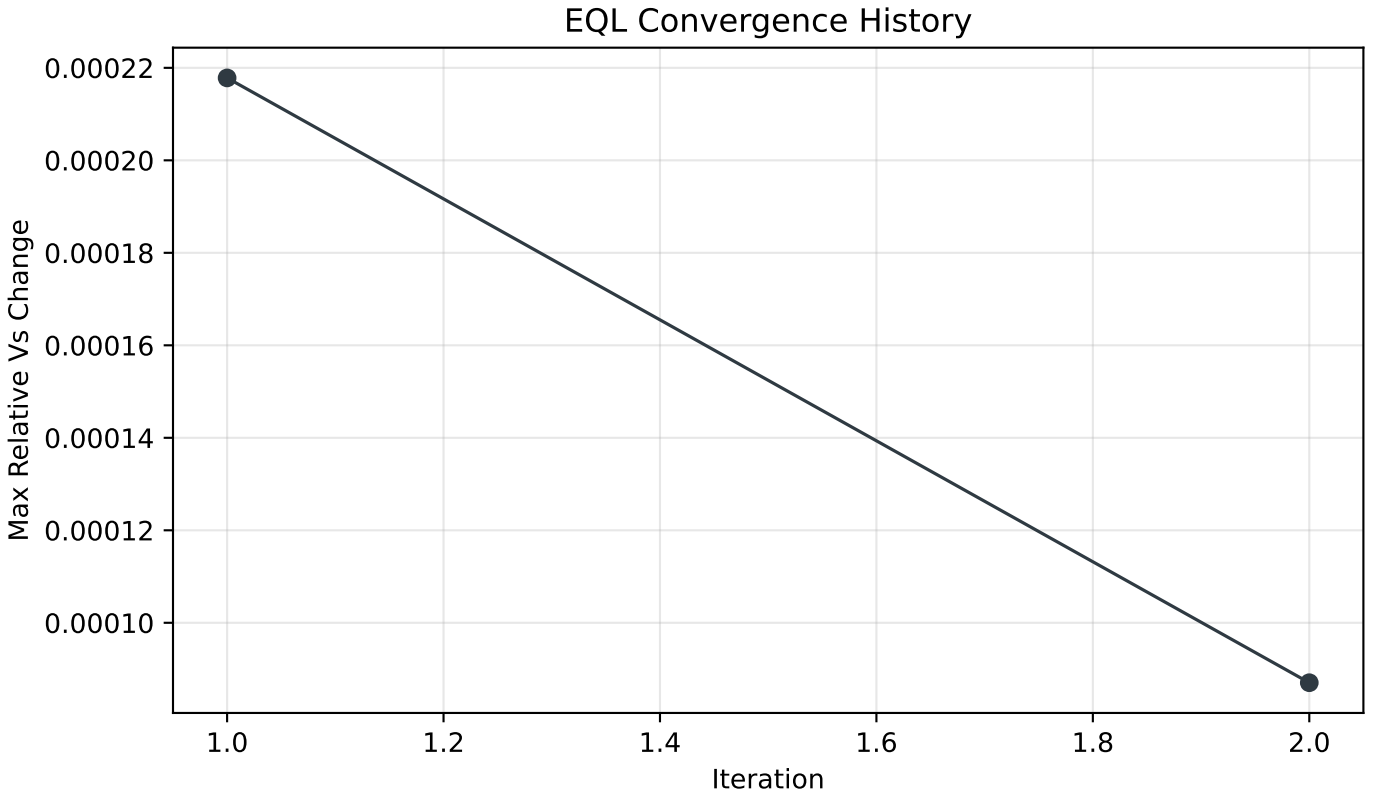
9. Reproducibility

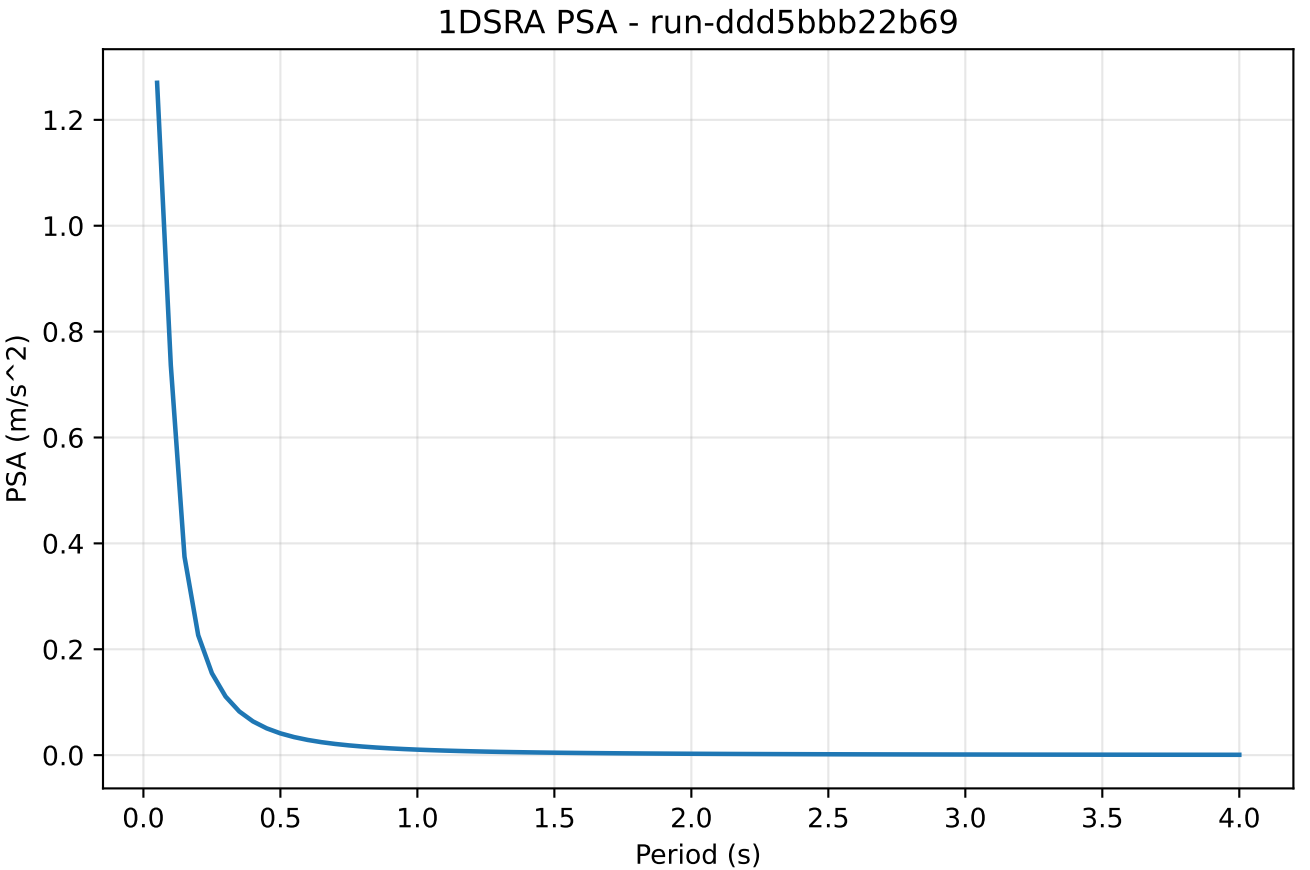
Rebuild the pack with: python scripts/build_validation_pack.py

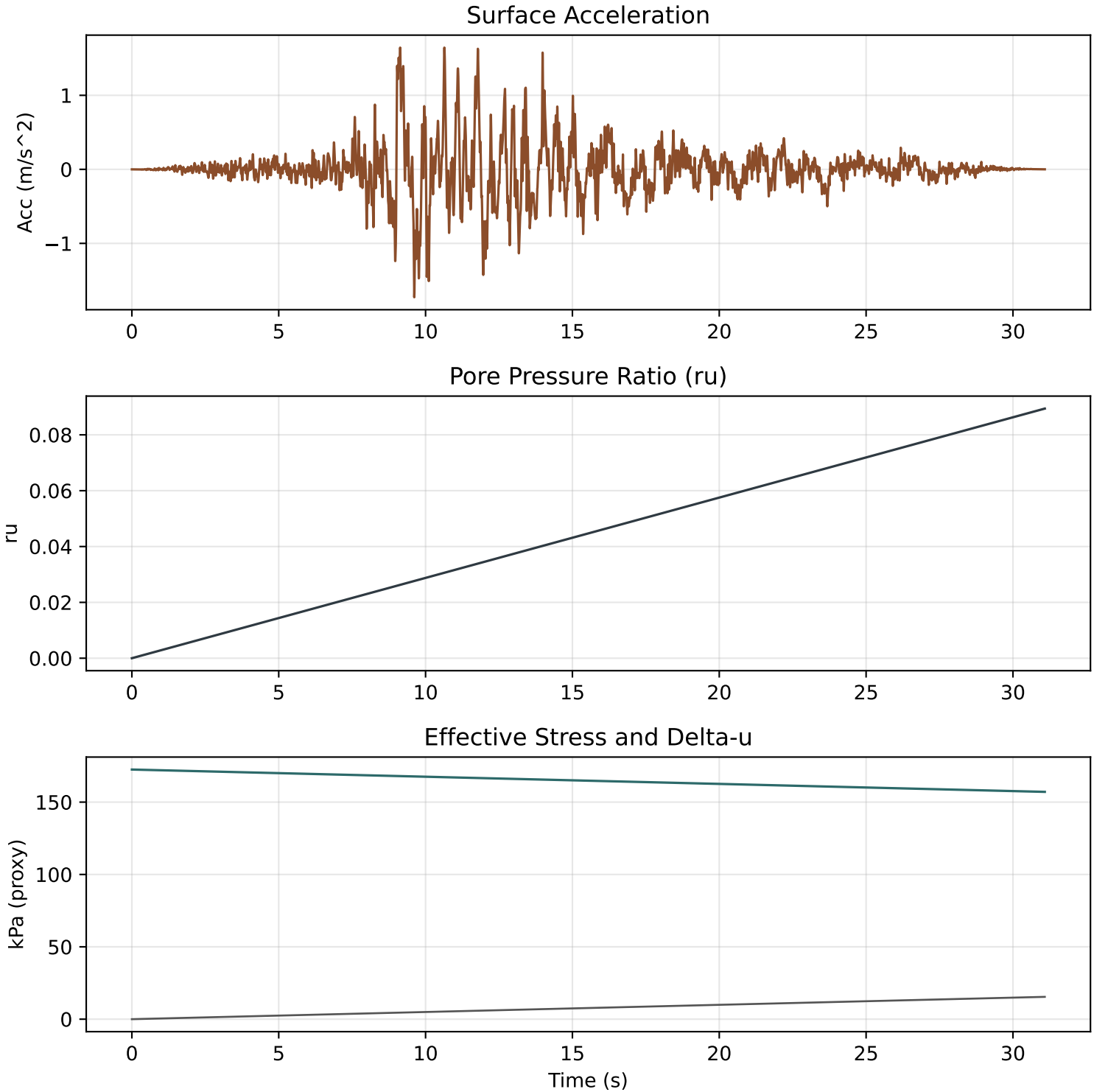

StrataWave PSA - run-2c32ae6cc181
PSA (m/s^2)
0.10
0.08
0.06
0.04
0.02
0.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Period (s)

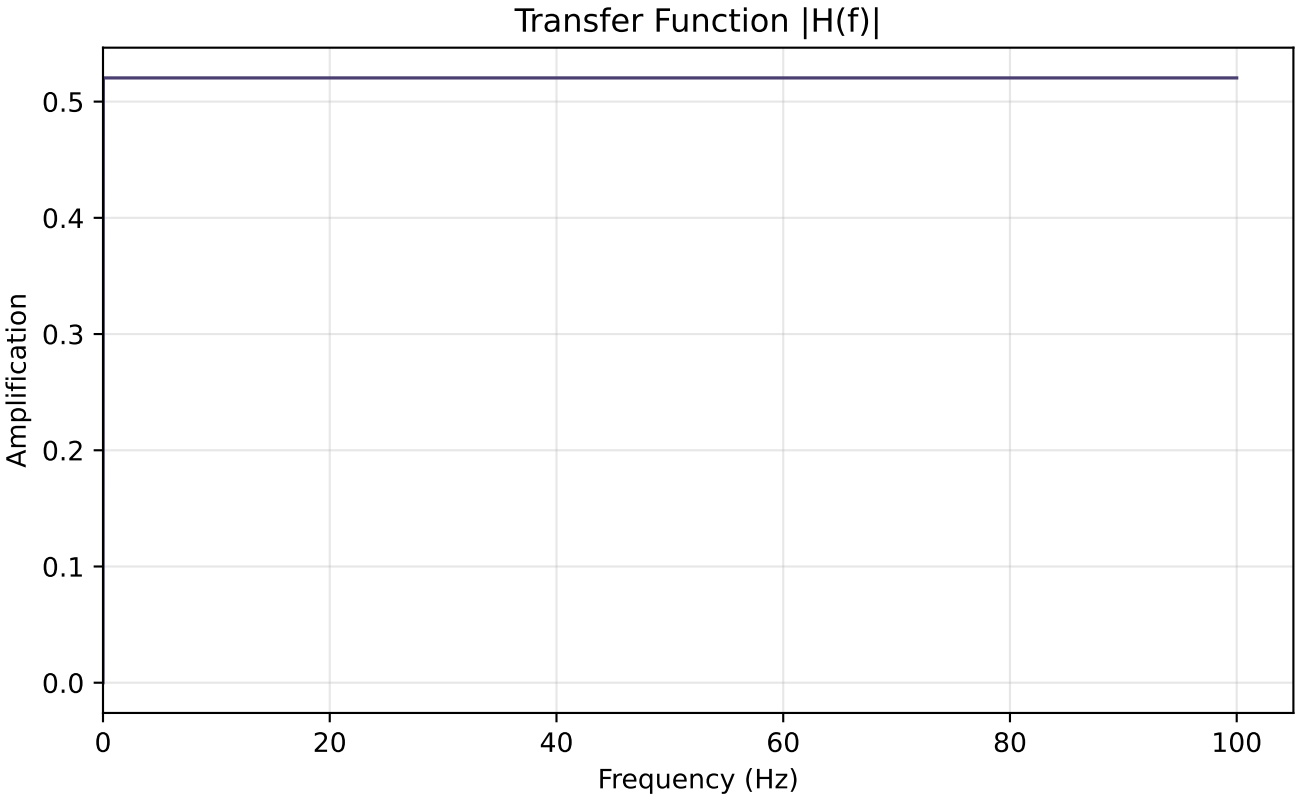

Surface Acceleration
Acc (m/s^2)
0.04
0.02
0.00
−0.02
−0.04
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Pore Pressure Ratio (ru)
ru
0.04
0.02
0.00
−0.02
−0.04
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Effective Stress and Delta-u
kPa (proxy)
150
100
50
0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Time (s)

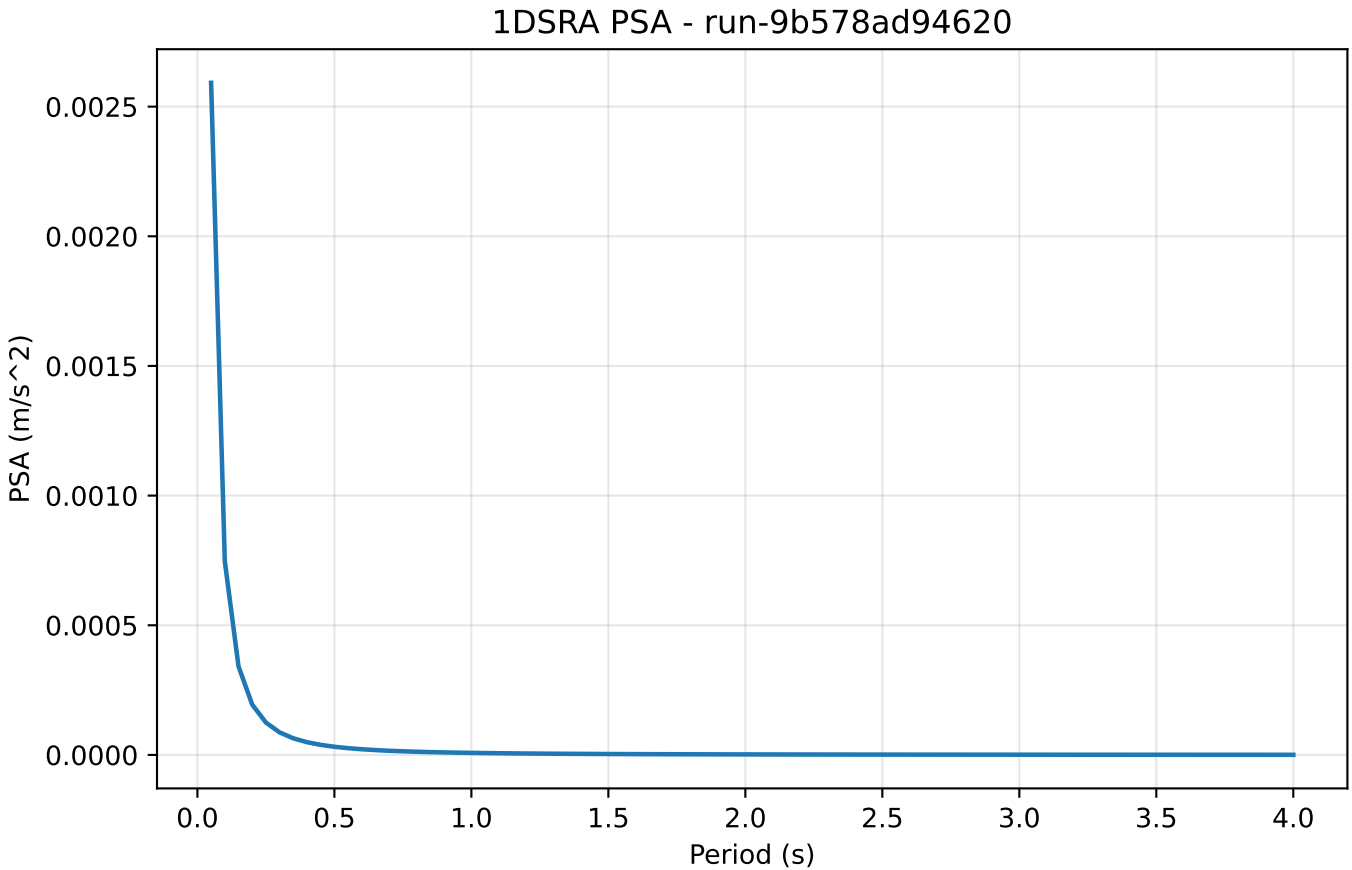

Transfer Function |H(f)|
Amplification
Frequency (Hz)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
20
40
60
80
100

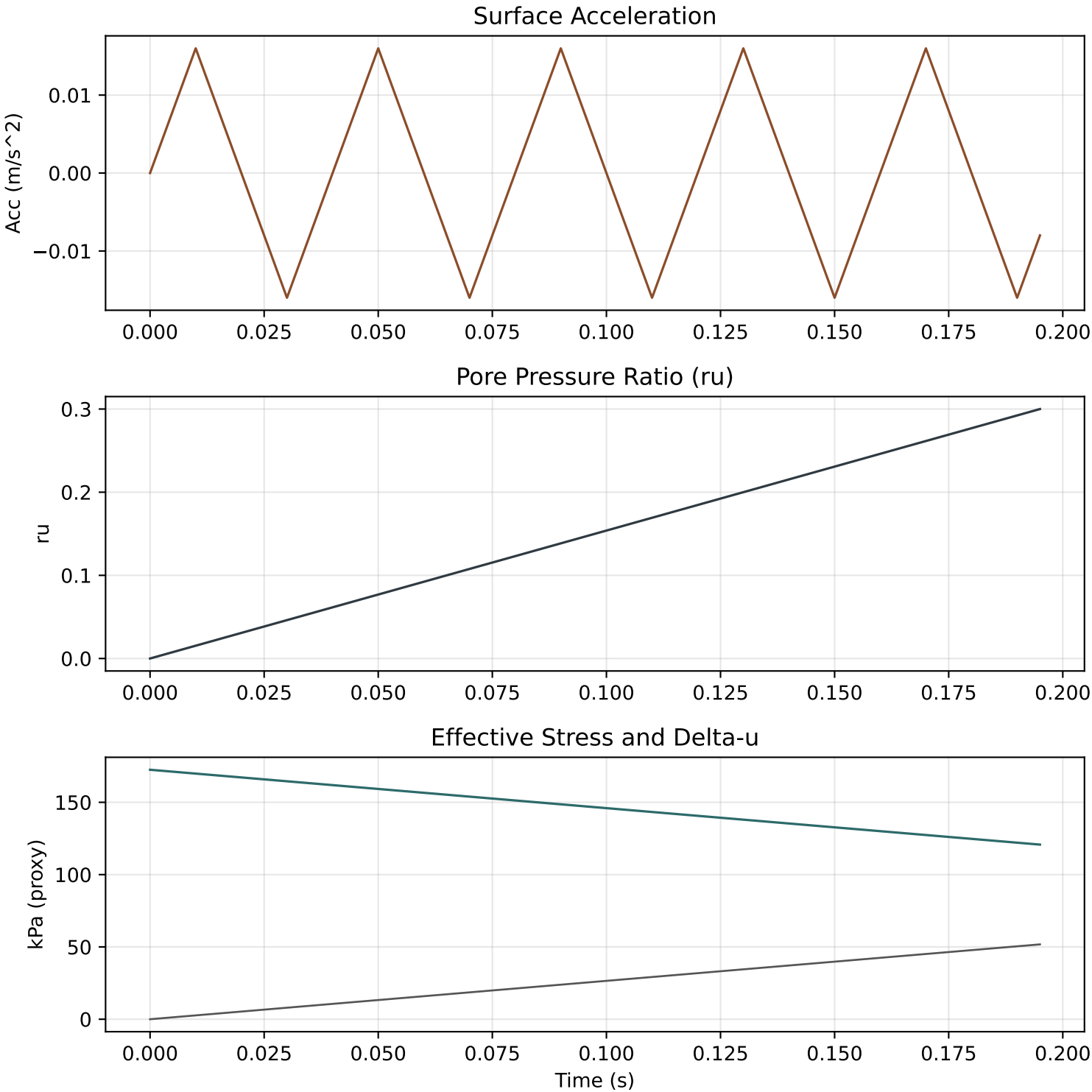

EQL Convergence History
Max Relative Vs Change
0.00022
0.00020
0.00018
0.00016
0.00014
0.00012
0.00010
1.0
1.2
1.4
1.6
1.8
2.0
Iteration

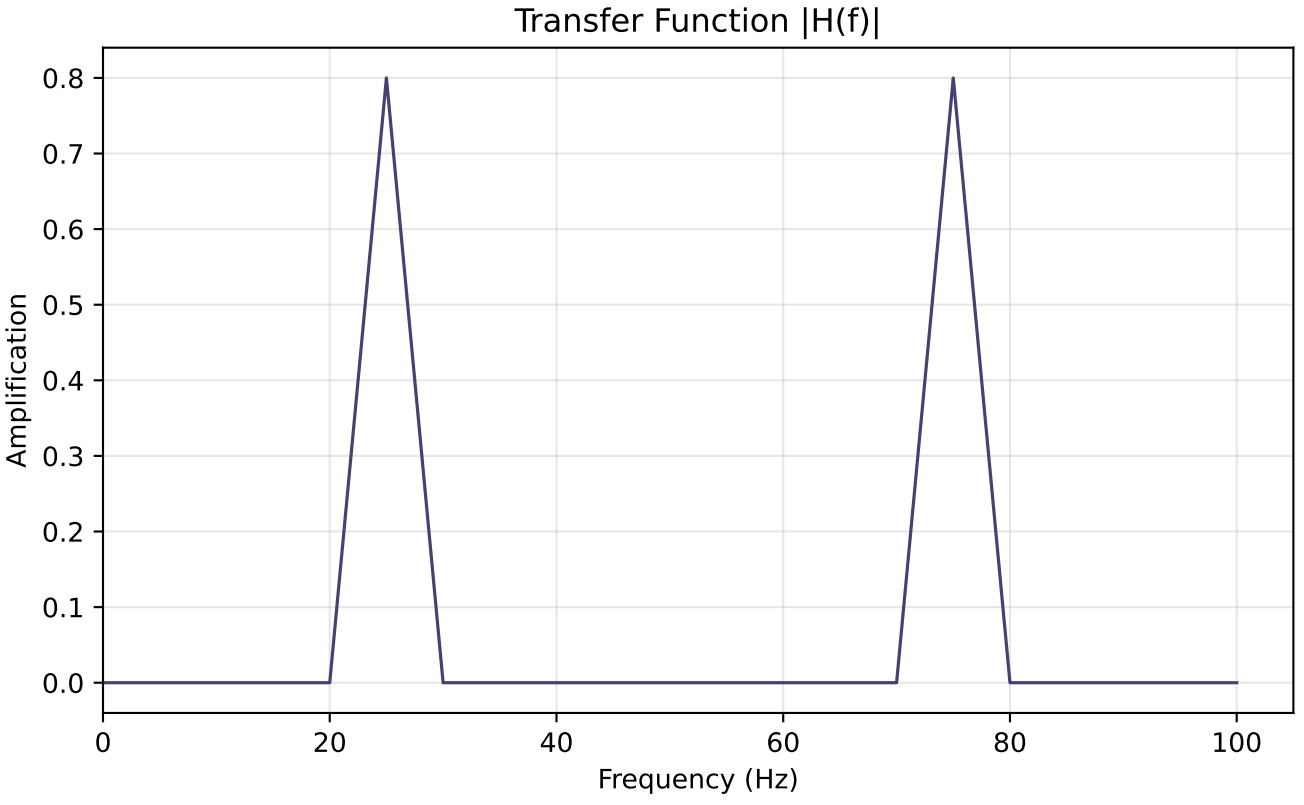

1DSRA PSA - run-ddd5bbb22b69
PSA (m/s^2)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Period (s)


Surface Acceleration
Acc (m/s^2)
1
0
−1
0
5
10
15
20
25
30
Pore Pressure Ratio (ru)
ru
0.08
0.06
0.04
0.02
0.00
0
5
10
15
20
25
30
Effective Stress and Delta-u
kPa (proxy)
150
100
50
0
0
5
10
15
20
25
30
Time (s)


Transfer Function |H(f)|
Amplification
0.5
0.4
0.3
0.2
0.1
0.0
0
20
40
60
80
100
Frequency (Hz)


1DSRA PSA - run-9b578ad94620
PSA (m/s^2)
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Period (s)


Surface Acceleration
Acc (m/s^2)
0.01
0.00
−0.01
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Pore Pressure Ratio (ru)
ru
0.3
0.2
0.1
0.0
Effective Stress and Delta-u
kPa (proxy)
150
100
50
0
Time (s)


Transfer Function |H(f)|
Amplification
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
20
40
60
80
100
Frequency (Hz)